



• Before 2017, lithium-ion batteries existed in the market but the quantum of these batteries has vastly increased from the time electric mobility came into the picture.

• The annual lithium-ion battery potential from EVs is likely to reach 130 gigawatt hours (GWh) in India by FY2030, from 5GWh as of FY2022.

• At present, EVs account for a 30% share of the overall lithium-ion batteries' deployment in India. By 2030, the share is expected to increase to 60%.

• Within EVs, electric 2-wheelers will be the major segment which will hold a majority share of over 80% by 2030. The electric 3-wheelers will also have a substantial share.

• On the basis of this demand for lithium-ion batteries, their recycling market potential in India is expected to cross 48GWh by 2030.

• The lithium-ion batteries can be recycled as well as refurbished for reuse.

• From the perspective of recyclers, the majority of scrap coming out in India is from mobile phone batteries. The electric 2-wheeler batteries have started coming in, but these have NMC (nickel, manganese, and cobalt) cells.

• In the case of electric 2-wheelers and electric 3-wheelers, nearly 75% of such batteries are being recycled and the rest 25% are estimated to be used for a second life, like behind-the-meter (energy production and storage systems that directly supply homes and buildings with electricity) applications.

• In the case of e-cars and e-buses, second-life usage is as high as 60%.

• The battery recycling process entails pre-treatment, which constitutes the thermal process and mechanical processes. Then comes post-treatment, for which two technologies are deployed: pyrometal-

RECOVERY TARGETS FOR BATTERIES

Type of battery	Recovery target for the year in percentage		
	2024-25	2025-26	2026-27 and onwards
Portable	70	80	90
Automotive	55	60	60
Industrial	55	60	60
Electric vehicle	70	80	90

Note. Maximum recovery target is subject to the percentage of non-recoverable hazardous material content in the battery. It would mean the reduction of recovery target by the same percentage of the hazardous material present in the waste battery.

Recovery of minimum percentage target is the percentage of the total weight of all recovered materials out of the dry weight of the battery. Recyclers shall be mandated for minimum recovery of battery materials, as mentioned in the Table, under Battery Waste Management Rules, 2022.

lurgy and hydrometallurgy. In India, hydrometallurgy is the most used. Direct recycling is also possible, which is more favourable for LFP (lithium iron phosphate).

• Lithium nickel manganese cobalt (NMC) oxide batteries' chemistry provides higher energy density, low weight, etc, while LFP provides higher safety and better cycle life. We are moving more towards LFP but there isn't much to recover here.

• We are going through this technological evolution. Some other chemistries will also come soon. The suitability for the recycling is going to be determined by the original equipment manufacturers (OEMs) themselves.

The challenges

Here's a list of challenges posed by the increasing sale of EV batteries:

• Limited resource availability is a challenge. As the government of India is pushing for local manufacturing of lithium cells, demand for raw materials is expected to grow significantly. But, currently, there are no local resources for such rare metals in India.

• A major challenge is related to the material coming out through the first level of recycling. It is very difficult to check if the black mass generated out of recycling would have the same specifications which are being promoted and provided

by these mechanical recycling players.

• The idea of tapping the B2C segment by recyclers is also logistically inconceivable. Only battery manufacturers can tap the market through extended producer requirement norms, if imposed strictly.

• Geopolitical risk, war, and the pandemic are creating additional threats that are disrupting the supply-chain. The global supply-chain, especially China's, is resulting in a long lead time for raw material deliveries.

• The most important challenge is the high cost, huge upfront costs involved in setting up recycling plants, and high processing costs that are becoming the main deterrents.

• We do not yet have enough batteries that have actually gone through the cycle life to really talk about the critical mass.

• Unlike a traditional lead-acid battery, a lithium-ion battery, if not fully discharged before it goes for black mass, is bound to create problems. Being handled by the informal sector is a disaster in making.

• The process of recycling EV batteries is still in the learning phase. Dismantling and recycling of batteries present a huge task in terms of setting up facilities for the production of the black mass or derivatives, such as cobalt sulphate, nickel sulphate, and lithium carbonate.

• In terms of channelisation, battery recycling is still being handled by the large informal sector, which was earlier handling the e-waste, because significant battery supplies are from smaller devices like smartphones, other gadgets, and power tools.

• There are multiple channels before a battery reaches the end recycler. There are a lot of leakages in the system, which make it an inefficient process.

• There are no guidelines from the government for second-life bat-